

Turkish Airline flight 1951

Flight model: Boeing 737
Date: February 25, 2005

From: Istanbul , Turkey
To: Amsterdam, Netherlands

Boeing 787 800 series is designed to fly long flights at higher altitudes. The flight is said to have simple system and improved wing.

Problem

During landing approach, just few miles away from the runway the flight had just dropped from the sky on the ground. There was no fire after crash. The plane had reported no emergency calls or call for system failure. The plane wreckage was not scattered. There was plenty of fuel in the plane tanks.

Investigation

Passengers report that at the landing approach there was a mild turbulence which leads to the possibility of a **microburst**. Microburst is a phenomena where a gust of air slams the plane to the ground whose possibility was later rejected. When checking the flight cockpit voice recorder, there is a warning that indicates the landing gear. When checking the flight data recorder, one of the instrument that measures -8 which means that flight is on ground but in reality the flight is in mid air (2000 feet). The Boeing 737 has 2 altimeter, **Pressure altimeter** measures the air pressure to measure height above sea level which seems to have worked properly. The other one is a **radio altimeter**, which is made up of 4 antennas, 2 sends signal to the ground and 2 reads the signals from the ground. Pressure altimeter are usually precise but not accurate. But, the later is more accurate. The captain side of the display displays reading from the radio altimeter and the first officers altimeter displays reading from the pressure altimeter. As the reading from the radio altimeter said the plane was on ground but the gear were not deployed lead to the warning signs. The ATC had asked to pilots to line with the runway to catch the **ILS** signals to follow the glide slope. The ATC had directed the flight such that the plane was higher than usual ILS approach which is usual practice at Amsterdam airport. Here the pilots would have to perform **Slam and dump approach** were in the flight drops to certain altitude to capture the right slope. After this the pilots receive a **stall warning**. Moments before the crash the engines were at idle as per the FDR. When further investigating, it was found out that when the flight was in mid air the plane was in **retard mode** which directs the plane to touch down and reduce the engine power to idle. It was found that the radio altimeter provides data to the auto throttle which in this case is faulty. It was found that the problem of going into retard mode at the time of landing approach was found in many of the 737. As per Boeing, the fault was not a safety hazard as the 737 gives multiple warning, but if not handled properly this fault can be catastrophic.

Changes that were brought after the incident

The Dutch safety board had recommended the Boeing to look into the faulty altimeter in the 737. A study says we have to design the automation in area where the individual doesn't do it well. The advancement in technology has lead to a situation where the systems are so complex that we are not able to properly talk to the systems and when a fault occurs the result would be catastrophic. This leads to the conclusion that the best technology should not replace the humans rather must with them. Airbus takes much decision making to the computers and Boeing wants the pilots to make decision with the data provided by the system.

Reference

1. A Crash That Made Boeing Change Their System— Mayday: Air Disaster
<https://www.youtube.com/watch?v=C9EI9oIucSU&t=57s>